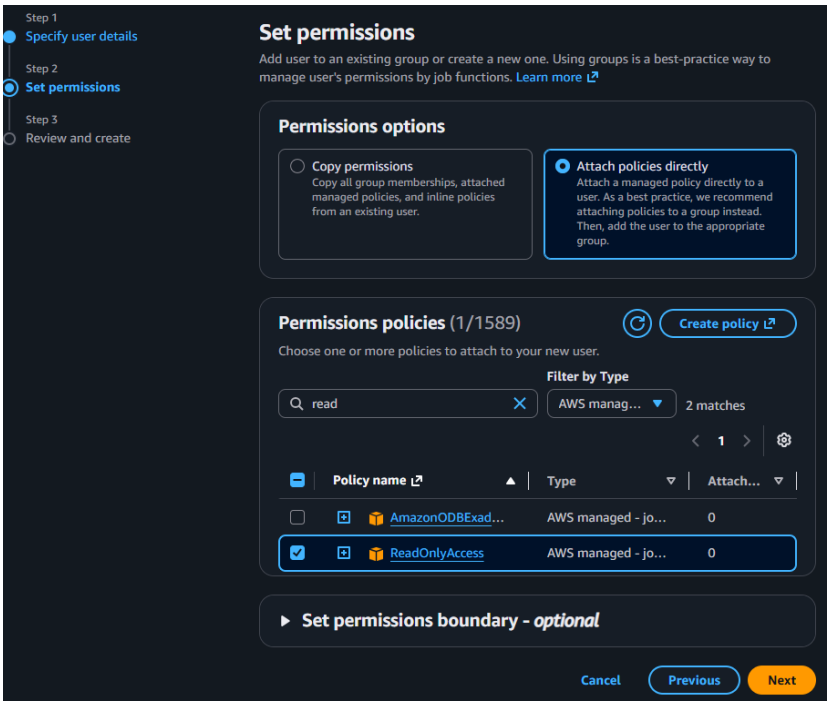
AWS IAM LAB TASKS

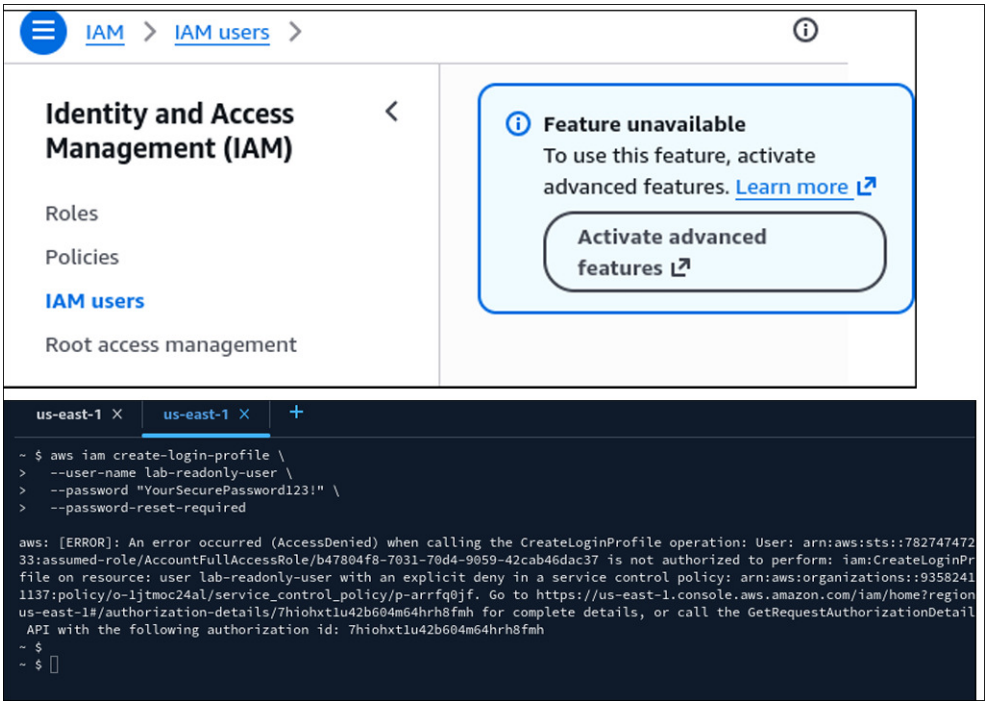
Submitted by: **Hrishinandan P**

Task 1: IAM User with Console Access

Result: User created with the ReadOnlyAccess policy. Console password setup was blocked by an organization Service Control Policy, so console login could not be tested.



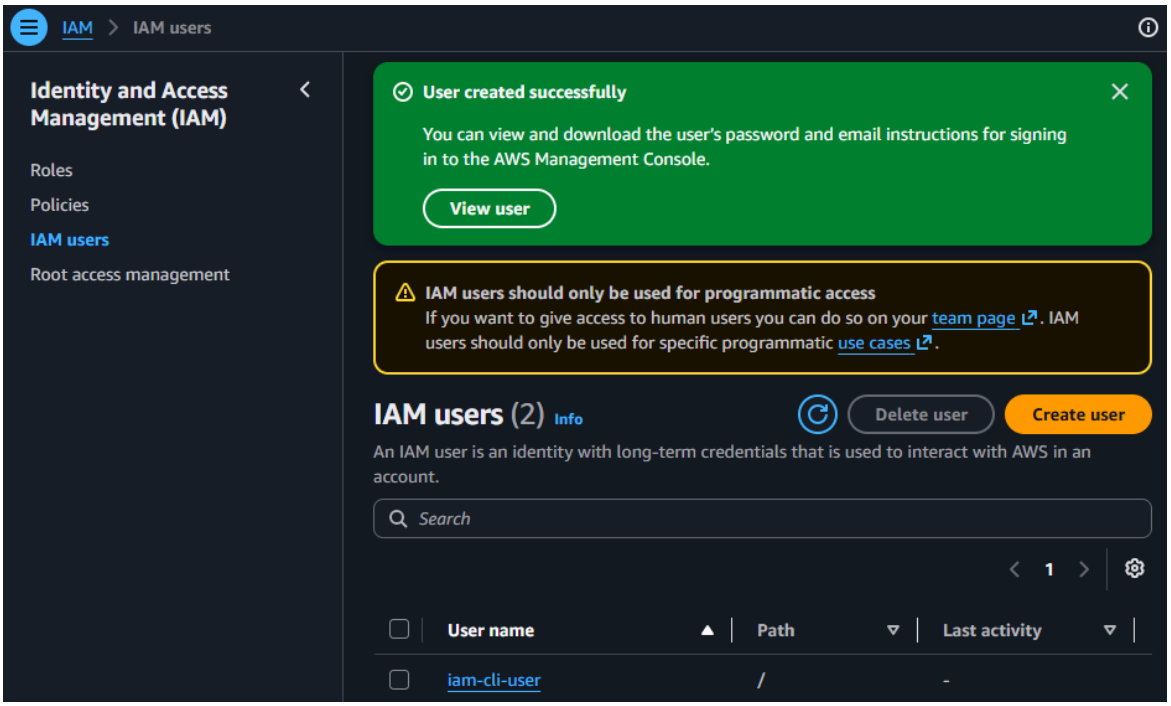
Screenshot 1: Attached the AWS managed ReadOnlyAccess policy to the IAM user.



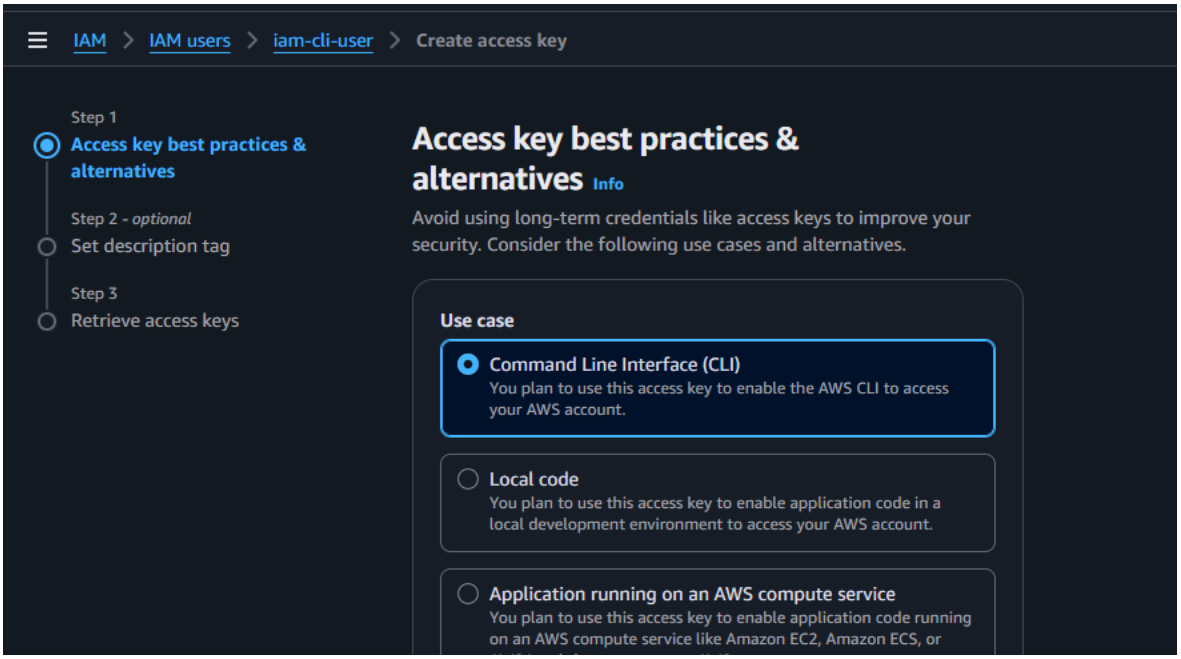
Screenshot 2: Console password setup showed "Feature unavailable"; the CLI command failed with AccessDenied due to an SCP explicit deny.

Task 2: IAM User without Console Access (AWS CLI)

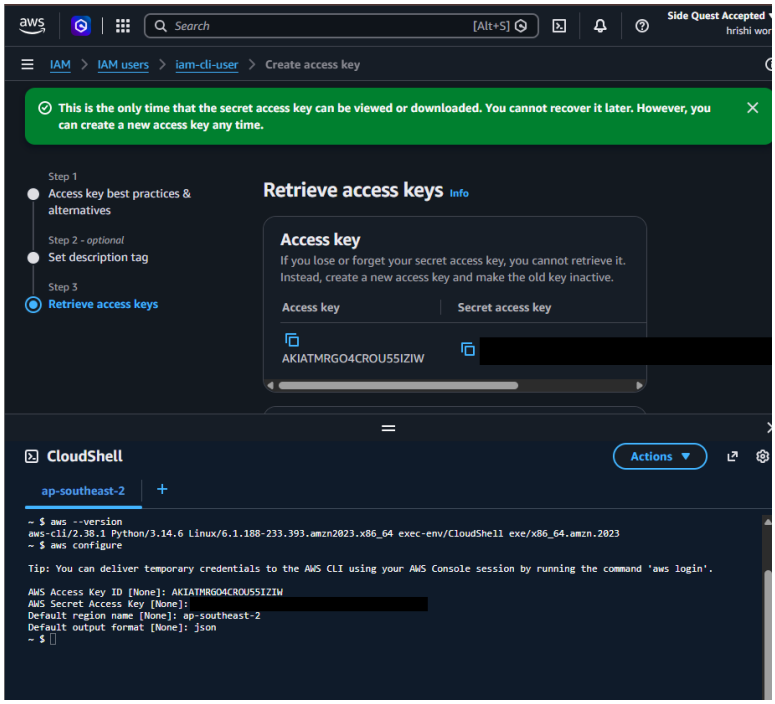
Result: CLI-only user created; access key worked and S3 buckets were listed.



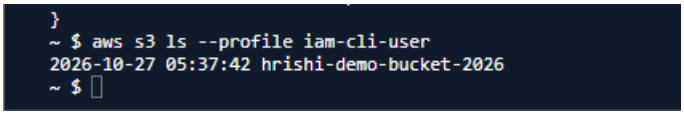
Screenshot 1: Created the IAM user iam-cli-user without console access.



Screenshot 2: Generated an access key with Command Line Interface (CLI) as the use case.



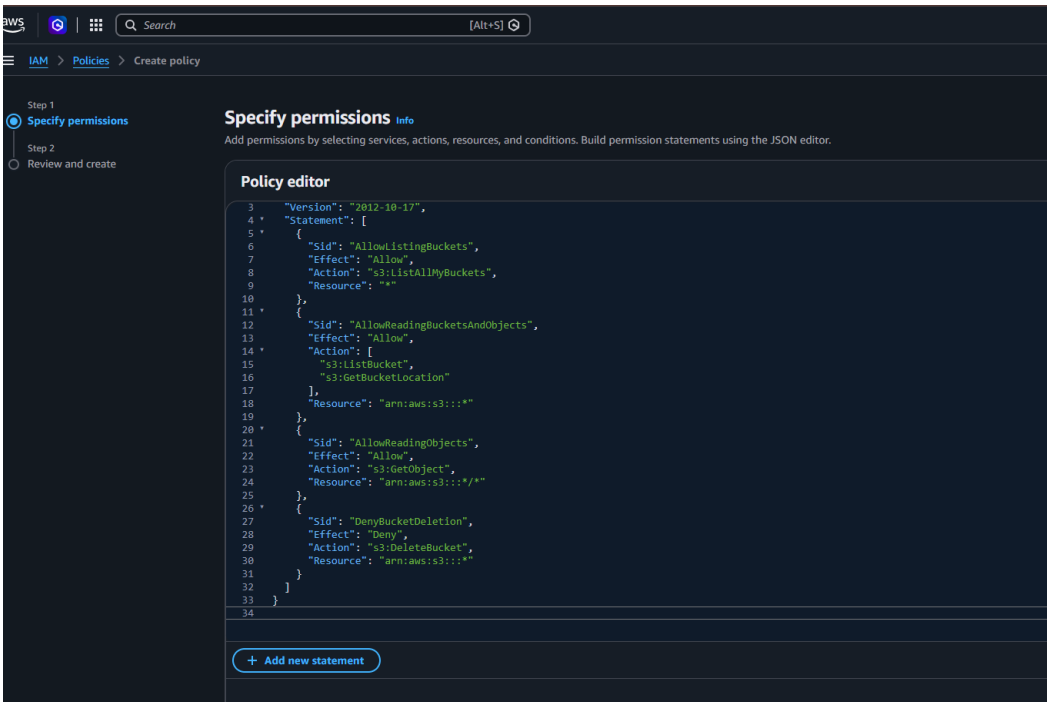
Screenshot 3: Retrieved the access key and configured the AWS CLI using aws configure (secret key hidden).



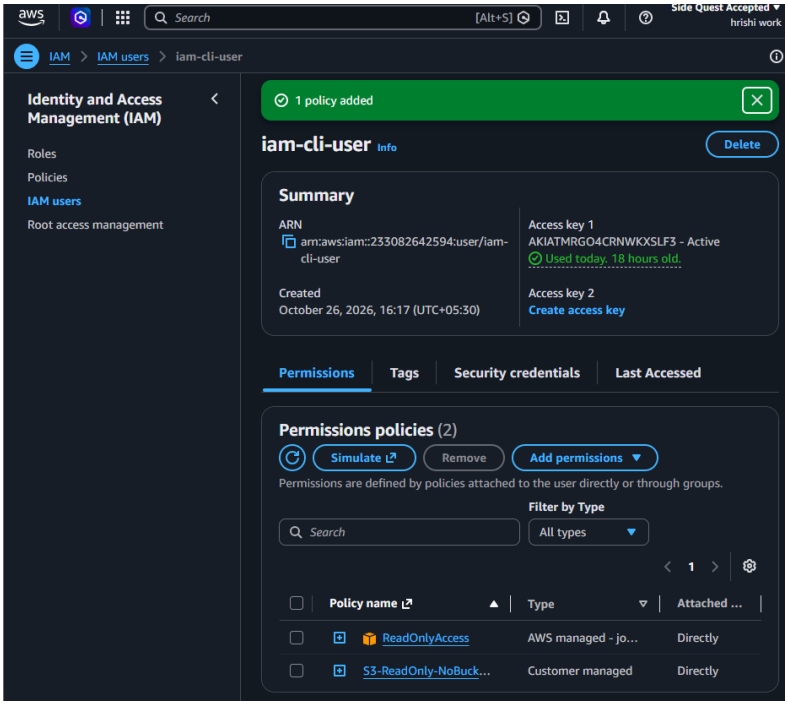
Screenshot 4: aws s3 ls succeeded and listed the bucket, confirming CLI authentication.

Task 3: Create a Custom IAM Policy

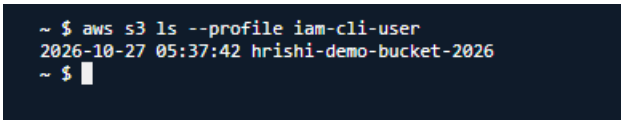
Result: Custom policy allows S3 listing and reading, and denies bucket deletion.



Screenshot 1: Custom policy S3-ReadOnly-NoBucketDelete with an explicit Deny on s3:DeleteBucket.



Screenshot 2: Policy attached to the user iam-cli-user.



Screenshot 3: Allowed: listing buckets succeeded.

```
~ $ aws s3api list-objects-v2 \
>   --bucket hrishi-demo-bucket-2026 \
>   --profile iam-cli-user
{
    "Contents": [
        {
            "Key": "Bmw Pic.avif",
            "LastModified": "2026-10-27T06:00:49+00:00",
            "ETag": "\"bbd7444a6a1174b543ab7217e5fae7c8\"",
            "ChecksumAlgorithm": [
                "CRC64NVME"
            ],
            "ChecksumType": "FULL_OBJECT",
            "Size": 8897,
            "StorageClass": "STANDARD"
        },
        {
            "Key": "index.html",
            "LastModified": "2026-10-27T06:01:06+00:00",
            "ETag": "\"2554e9555913879c34b8451d89adc6fe\"",
            "ChecksumAlgorithm": [
                "CRC64NVME"
            ],
            "ChecksumType": "FULL_OBJECT",
            "Size": 1126,
            "StorageClass": "STANDARD"
        }
```

Screenshot 4: Allowed: listing and reading objects in the bucket succeeded.

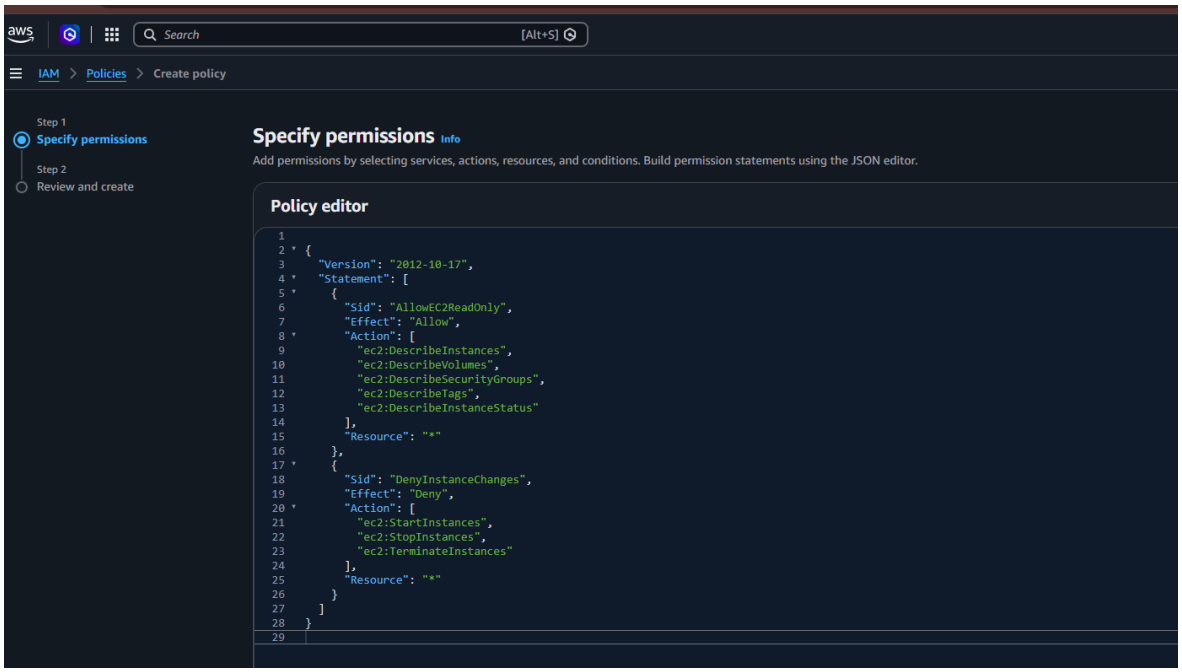
```
~ $
~ $ aws s3api delete-bucket \
>   --bucket hrishi-demo-bucket-2026 \
>   --profile iam-cli-user

aws: [ERROR]: An error occurred (AccessDenied) when calling the DeleteBucket operation: User: arn:aws:iam::233082642594:user/iam-cl
-user is not authorized to perform: s3:DeleteBucket on resource: "arn:aws:s3:::hrishi-demo-bucket-2026" with an explicit deny in an
identity-based policy: arn:aws:iam::233082642594:policy/S3-ReadOnly-NoBucketDelete
~ $
```

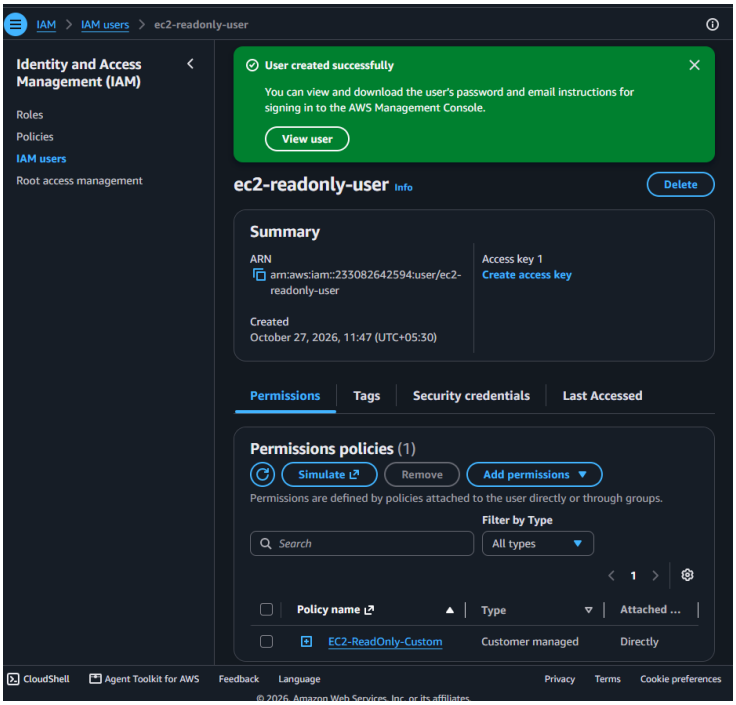
Screenshot 5: Denied: delete-bucket failed with AccessDenied due to the explicit deny in the policy.

Task 4: EC2 Read-Only Access

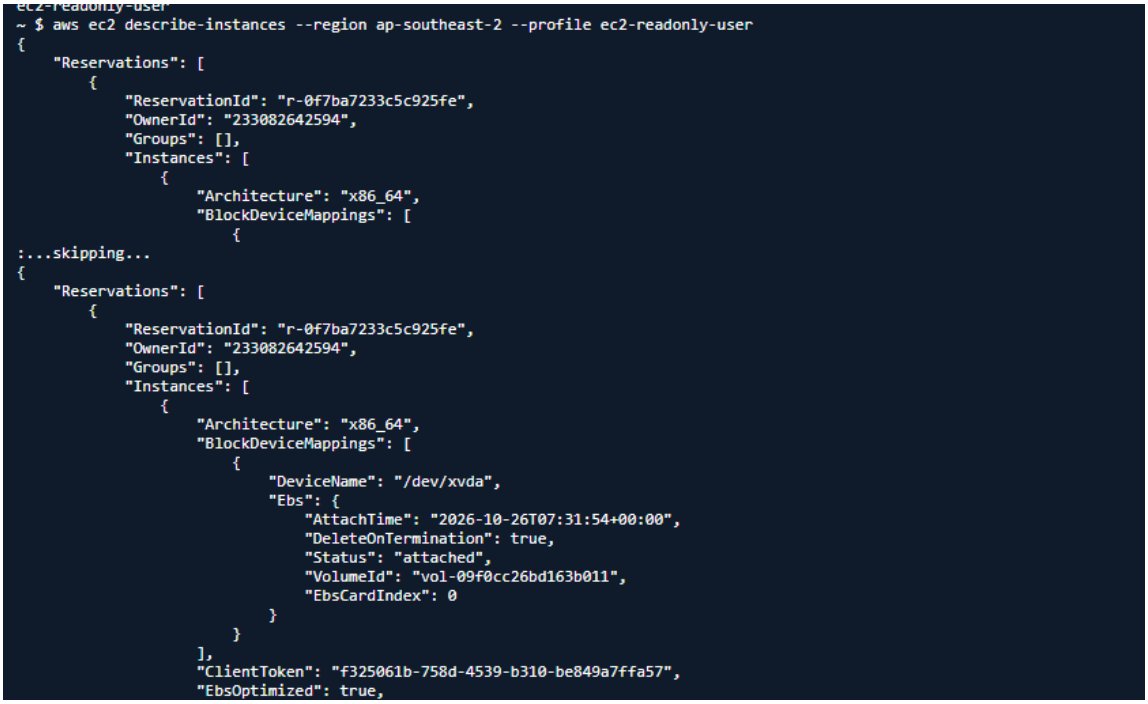
Result: User can view EC2 resources but cannot start, stop or terminate instances.



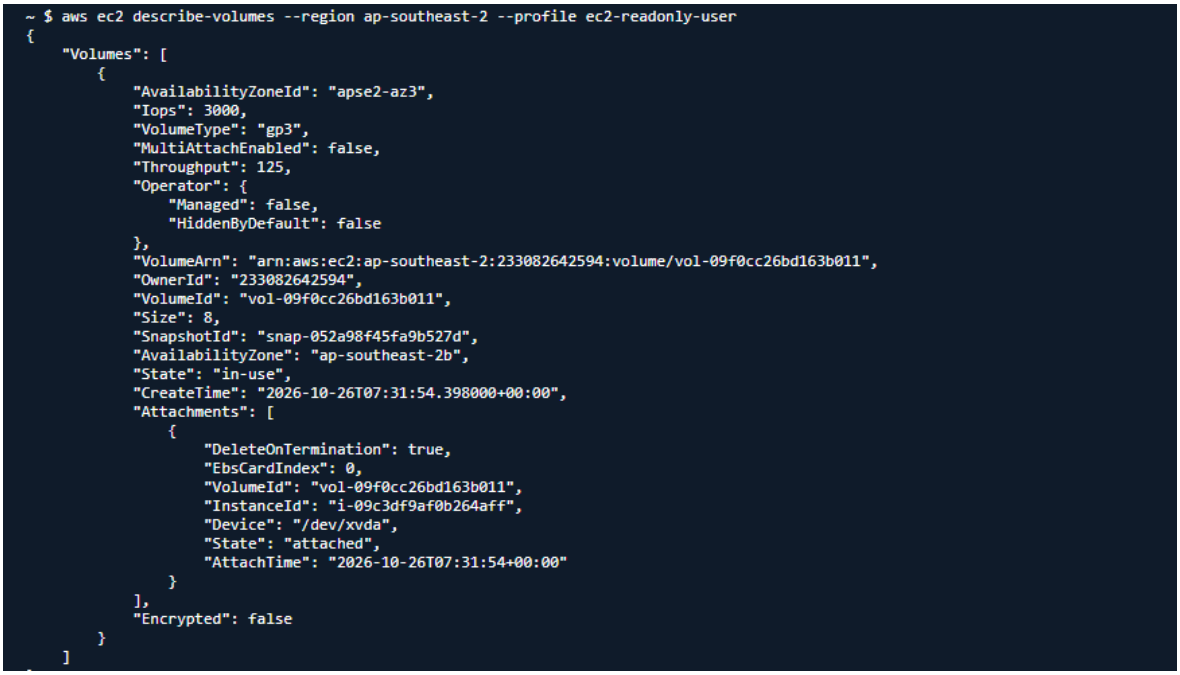
Screenshot 1: Custom policy EC2-ReadOnly-Custom: allows Describe actions, denies Start, Stop and Terminate.



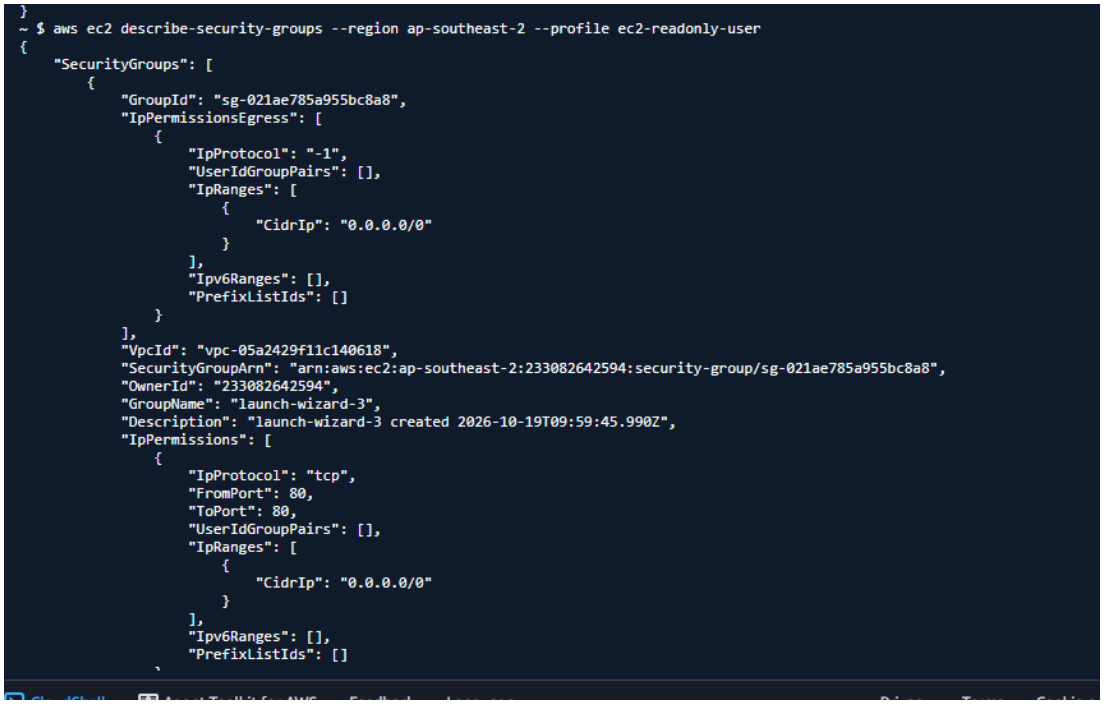
Screenshot 2: User ec2-readonly-user created with only this policy attached.



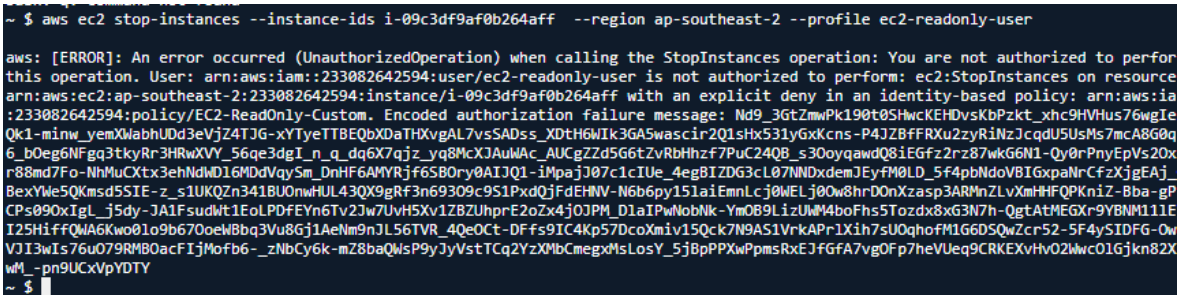
Screenshot 3: Allowed: viewing instances.



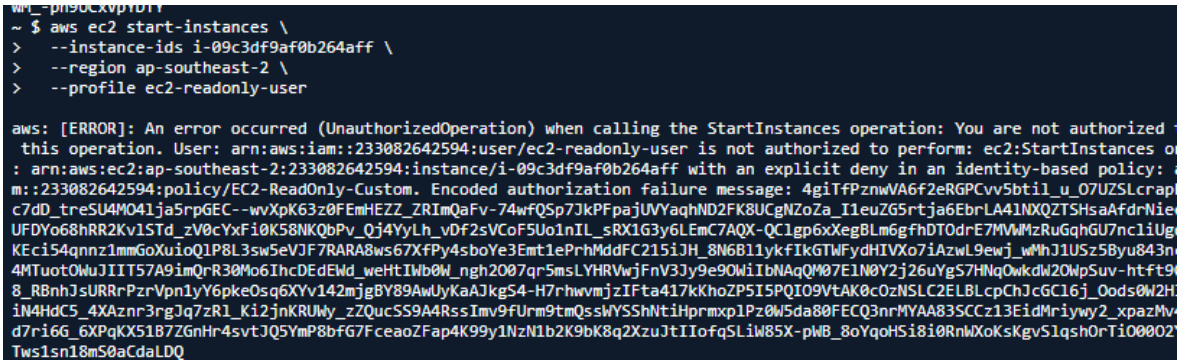
Screenshot 4: Allowed: viewing volumes.



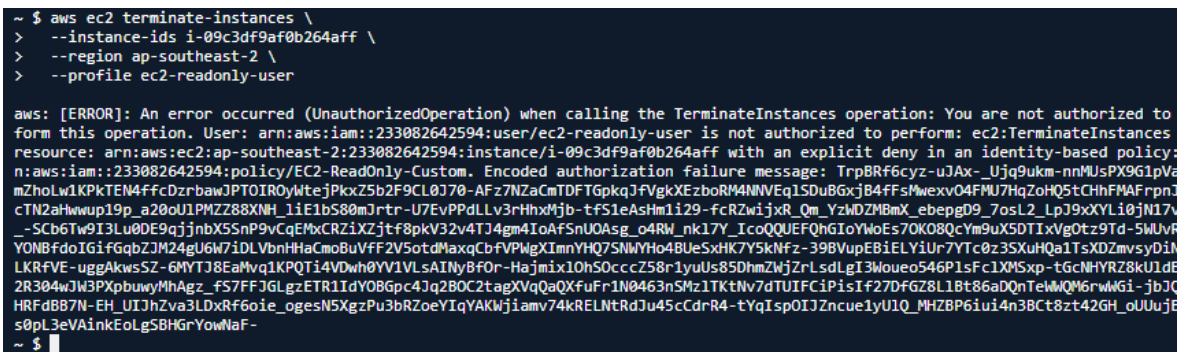
Screenshot 5: Allowed: viewing security groups.



Screenshot 6: Denied: stop-instances failed with UnauthorizedOperation.



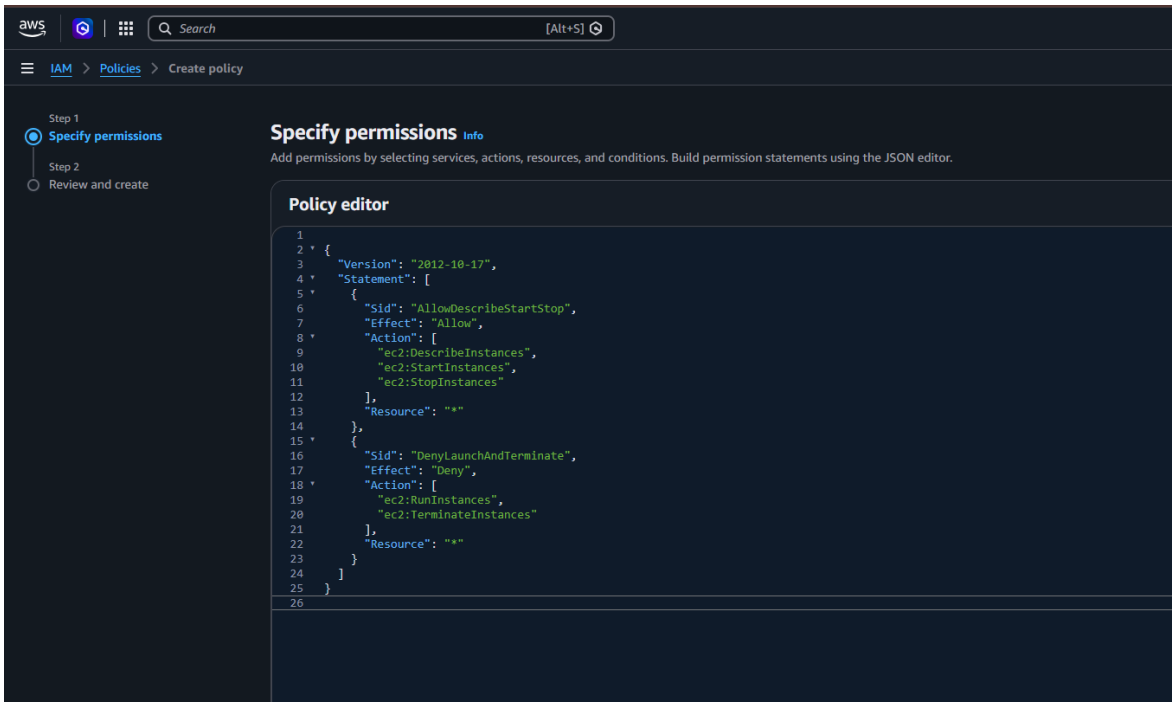
Screenshot 7: Denied: start-instances failed with UnauthorizedOperation.



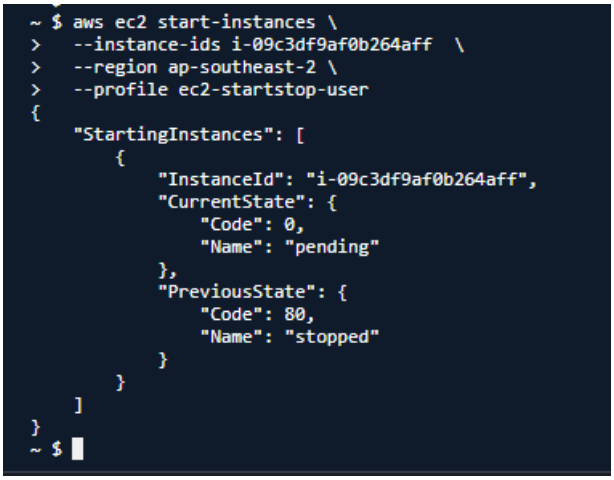
Screenshot 8: Denied: terminate-instances failed with UnauthorizedOperation.

Task 5: Start and Stop EC2 Only

Result: User can start and stop instances; terminating is denied.



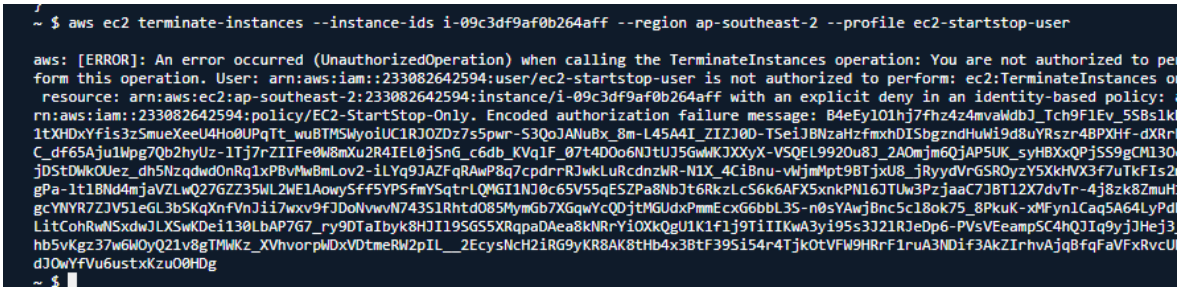
Screenshot 1: Custom policy allows DescribeInstances, StartInstances and StopInstances, and denies RunInstances and TerminateInstances.



Screenshot 2: Allowed: start-instances succeeded (stopped to pending).



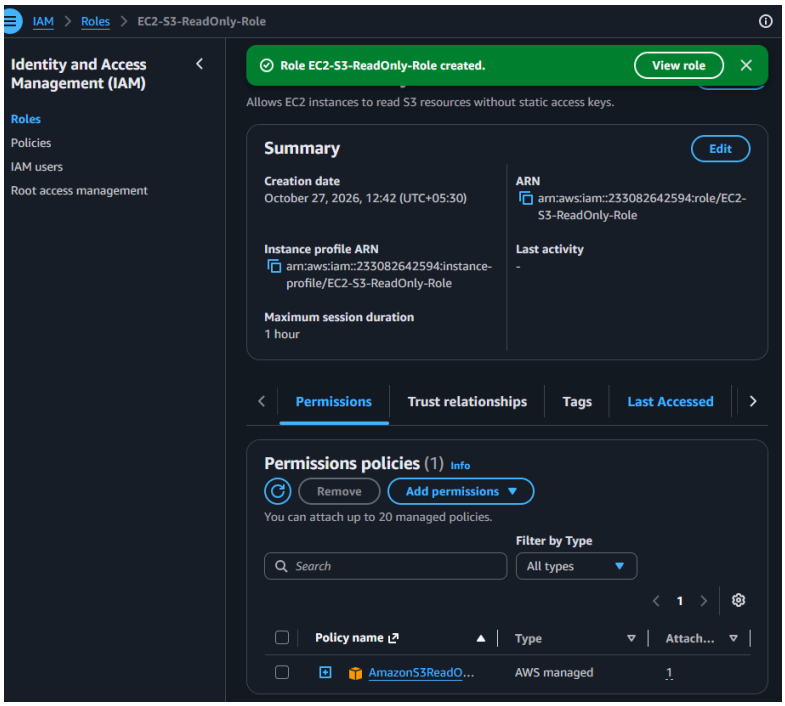
Screenshot 3: Allowed: stop-instances succeeded (running to stopping).



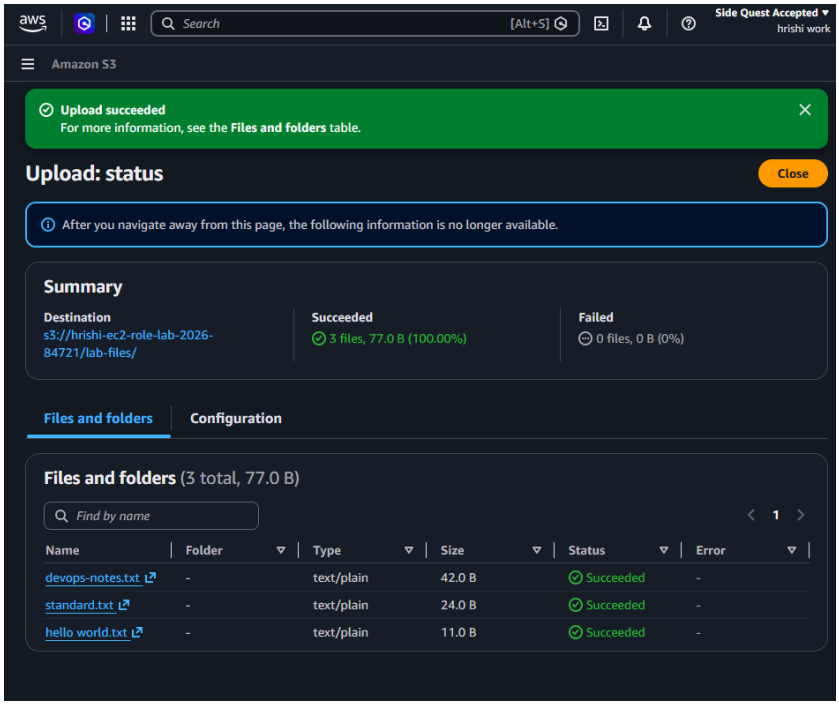
Screenshot 4: Denied: terminate-instances failed with UnauthorizedOperation.

Task 6: IAM Role for EC2

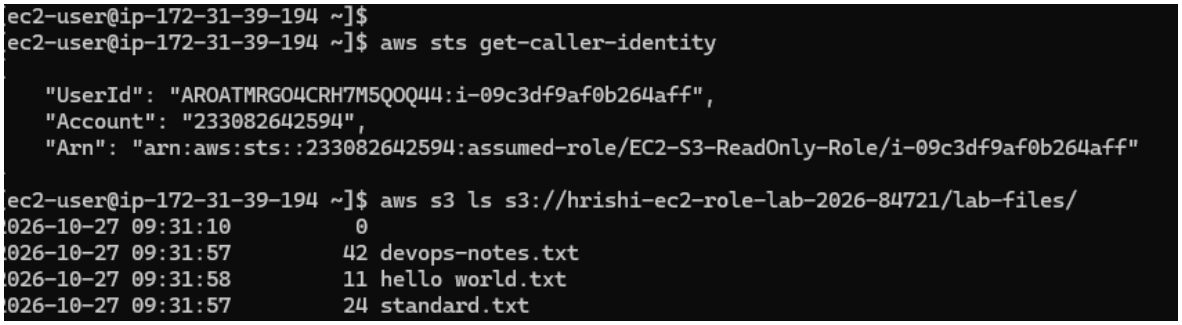
Result: EC2 instance accessed S3 through the IAM role without any access keys.



Screenshot 1: Created the role EC2-S3-ReadOnly-Role with the AmazonS3ReadOnlyAccess policy.



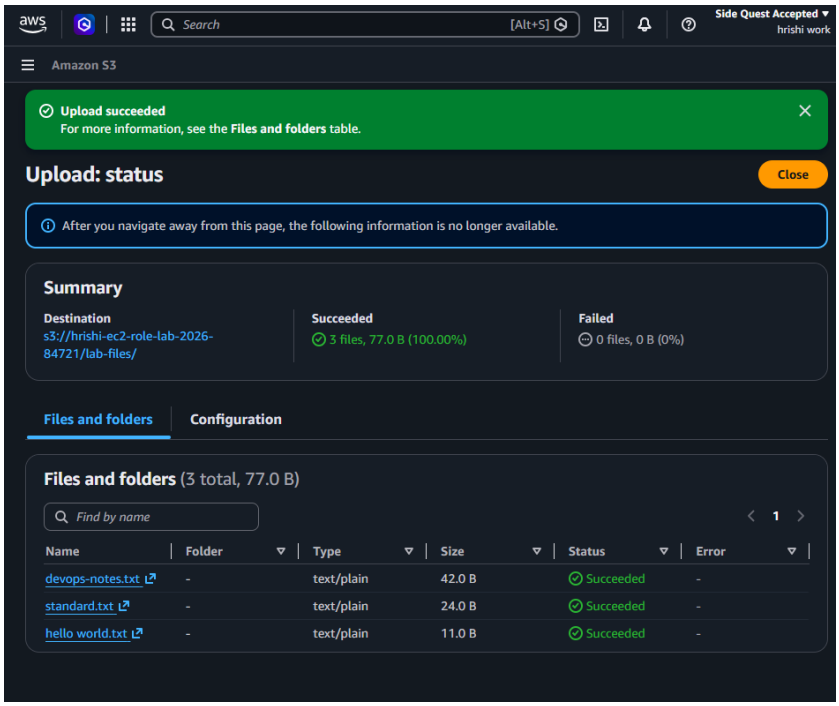
Screenshot 2: Created an S3 bucket and uploaded 3 sample files.



Screenshot 3: On EC2, STS shows the assumed role and S3 listing works without configuring keys.

Task 7: Access S3 from EC2 Using IAM Role

Result: Role allows listing and downloading S3 objects; uploads are denied.



Screenshot 1: S3 bucket created and sample files uploaded.

```
2026-10-27 09:31:57         24 standard.txt
[ec2-user@ip-172-31-39-194 ~]$ mkdir -p ~/s3-lab
[ec2-user@ip-172-31-39-194 ~]$ cd ~/s3-lab
[ec2-user@ip-172-31-39-194 s3-lab]$ aws s3 cp \
  s3://hrishi-ec2-role-lab-2026-84721/lab-files/devops-notes.txt \
  ./devops-notes.txt
download: s3://hrishi-ec2-role-lab-2026-84721/lab-files/devops-notes.txt to ./devops-notes.txt
[ec2-user@ip-172-31-39-194 s3-lab]$ cat devops-notes.txt
create a text file called devops-notes.txt[ec2-user@ip-172-31-39-194 s3-lab]$
[ec2-user@ip-172-31-39-194 s3-lab]$
```

Screenshot 2: Allowed: downloaded devops-notes.txt from S3 to the EC2 instance.

```
[ec2-user@ip-172-31-39-194 s3-lab]$ aws s3 cp \
  upload-test.txt \
  s3://hrishi-ec2-role-lab-2026-84721/lab-files/upload-test.txt
upload failed: ./upload-test.txt to s3://hrishi-ec2-role-lab-2026-84721/lab-files/upload-test.txt An error occ
essDenied) when calling the PutObject operation: User: arn:aws:sts::233082642594:assumed-role/EC2-S3-ReadOnly-
3df9af0b264aff is not authorized to perform: s3:PutObject on resource: "arn:aws:s3:::hrishi-ec2-role-lab-2026-
files/upload-test.txt" because no identity-based policy allows the s3:PutObject action
[ec2-user@ip-172-31-39-194 s3-lab]$ aws s3 ls s3://hrishi-ec2-role-lab-2026-84721/lab-files/
2026-10-27 09:31:10          0
2026-10-27 09:31:57         42 devops-notes.txt
2026-10-27 09:31:58         11 hello world.txt
2026-10-27 09:31:57         24 standard.txt
[ec2-user@ip-172-31-39-194 s3-lab]$
```

Screenshot 3: Denied: upload failed with AccessDenied on s3:PutObject; bucket still has only the original files.